Point-in-Time Obligation-Aware Retrieval Planning for Financial Question Answering

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Abstract

Financial questions are often underspecified as retrieval problems: a single question can require several entities, fiscal periods and filing types, while documents become available only at particular times. We introduce FinPlan-RAG, a non-oracle retrieval planner that represents these requirements as explicit filing obligations and maintains an auditable evidence state. The planner targets one unresolved obligation at a time, applies a point-in-time cutoff, and stops only when the obligation set is closed. This design is motivated by an identification gap: a relevance score is not, in general, sufficient to decide whether the evidence needed for a financial answer is complete. We formalize the gap with an observational-equivalence construction and show that query budget, temporal availability and counterevidence are separate variables in the decision problem. In a deterministic controlled benchmark with 20 seeds, seven missingness/conflict scenarios and 5,600 cases, FinPlan-RAGincreases closure from 0.101 to 0.366 relative to single-shot retrieval and uses 8.87 rather than 10 queries on average. Removing counterevidence increases closure to 0.511 but reduces counterevidence recall from 0.605 to 0.315 and increases overclaiming. Removing the cutoff produces future leakage in 50.2% of cases. These results are mechanism evidence rather than a real-world QA or state-of-the-art claim. We release a reproducible implementation and a tool/stdio adapter for common code agents, including Codex and Claude Code.

1 Introduction

Financial QA systems must align a question with an entity, a reporting period, and a filing that was actually available when the question was asked. A system can retrieve a highly relevant sentence and still fail because it omitted a second required filing, used a later revision, or ignored a limiting disclosure. Most RAG pipelines expose a ranked list but leave these obligations implicit.

Our research question is: *does an explicit obligation state improve the next retrieval and stopping decision under a fixed cutoff and query budget?* We make three bounded contributions. First, we formalize a financial retrieval obligation as the tuple (e, y, f, p) for entity, fiscal year, filing type and fiscal period. Second, we give a compact identification argument showing why a score-only planner cannot guarantee closure when two worlds share the same observed score but differ in an unobserved required filing. Third, we implement and challenge the resulting planner with controlled baselines, component ablations and a temporal leakage stress test. The maintained repository and the result hash are part of the evidence boundary; no number is filled in from an unverified external run.

2 Related work and research gap

Retrieval-augmented generation combines a retriever with a reader (Lewis et al., 2020). Dense and sparse retrievers optimize relevance or lexical coverage, while temporal QA benchmarks expose long-range and multi-hop dependencies (Maharana et al., 2024; Wu et al., 2024). These systems provide useful coverage controls but do not by themselves represent which filings are still required. Financial QA adds a stronger constraint: the answer must be supported by information available at a declared cutoff.

The gap is therefore not that existing rankers are inaccurate in every case. It is that a scalar relevance score is an incomplete state for a stopping rule. Following the lifecycle-aware separation used by SQCAD, we keep candidate proposal, evidence qualification and reader generation distinct; FinPlan-RAG specializes that separation to financial filings and point-in-time obligations. Our hypothesis is deliberately falsifiable:

> H1: obligation-aware state planning improves closure and query efficiency under fixed information and budget, while explicit counterevidence and point-in-time filtering trade coverage for lower overclaiming and lower future leakage.

3 Identification gap and theory

Let $O(q) = \{(e, y, f, p)\}$ denote the obligations parsed from question q and let $D_{\leq c}$ be documents with availability timestamp no later than cutoff c. A planner observes a history h_t and chooses a query target a_t from the unresolved obligations. Let $S(h_t)$ be any score-only summary used by a baseline.

Definition 1 (Closure)**.** *The evidence state is closed if every obligation with a known document id is covered by at least one retrieved, point-in-time document and all required slots have a resolved vote. Closure is a process property; it is not answer correctness.*

Proposition 1 (Score insufficiency)**.** *Suppose there exist two latent worlds w_+ and w_- with the same observed history and score $S(h_t)$, but with opposite optimal actions for an unresolved obligation. For any score-only randomized rule that chooses the positive action with probability p, its worst-world regret is at least*

$$\max\{(1-p)m_+, pm_-\} \geq \frac{m_+ m_-}{m_+ + m_-},$$

where m_+ and m_- are the two action margins.

Proof. In w_+, taking the negative action incurs at least m_+ and occurs with probability $1-p$; in w_-, taking the positive action incurs at least m_- and occurs with probability p. Balancing the two lower bounds gives the stated harmonic-mean floor. □

Theorem 1 (Obligation-state value decomposition)**.** *Let $x_t = (U_t, E_t, c)$ contain the unresolved obligation set, evidence state and cutoff, and let a_t be either a targeted retrieval action or stop. If the one-step utility is $g(x_t, a_t)$ and the transition/observation kernel is $K_a(dx_{t+1}, do_{t+1} \mid x_t)$, then the finite-horizon action value satisfies*

$$Q_t(x, a) = \mathbb{E}[g(x, a)] + \gamma\, \mathbb{E}_{K_a} V_{t+1}(x'), \qquad V_t(x) = \max_a Q_t(x, a).$$

For two actions a, b, the lifecycle contrast is

$$\Delta_t(x) = Q_t(x, a) - Q_t(x, b) = \Delta_t^{\text{immediate}}(x) + \gamma \Delta_t^{\text{continuation}}(x).$$

Consequently, a score summary $S(x)$ can support an optimal score-only policy only if $\Delta_t(x)$ is measurable with respect to $S(x)$; otherwise an observation-equivalent pair with different signs of Δ_t exists.

Proof. The first identity is the Bellman recursion obtained by conditioning on the next state and observation. Subtracting the recursion for b from that for a gives the contrast decomposition. If Δ_t is not measurable with respect to S, there are x_+, x_- with equal score and opposite contrast signs; applying the score-insufficiency proposition yields a strictly positive worst-world regret for every score-only rule. □

The construction is financial rather than purely semantic: w_+ contains a limiting disclosure or a future revision that changes the required filing, whereas w_- does not. Both can share the same top-ranked passage. Thus the failure is at the estimand/state layer, not only a reader error.

Theorem 2 (Point-in-time safety)**.** *If the planner only admits chunks with available_at $\leq c$ into the evidence state, then any answer context emitted by the planner contains no future document. If the cutoff predicate is removed, a future document can be consumed whenever it has higher retrieval relevance than all visible hits.*

Proof. The first claim follows directly from the retrieval predicate. For the second, construct a case with a later chunk whose relevance is strictly larger than all chunks available by c; the maximum-score selection consumes the later chunk. □

4 FinPlan-RAG method

4.1 Obligation parsing

The parser maps "Q2 FY 2024", "first half of FY 2024" and annual language to normalized period requests. Annual Q4 requests map to a 10-K/FY obligation; quarterly requests map to 10-Q/Q1–Q3. The resolver intersects these signatures with visible filing metadata and emits stable, sorted obligations.

4.2 Obligation-aware retrieval

Given question q, obligation $o = (e, y, f, p)$, and unresolved set U_t, the planner forms a targeted query $q \oplus o$ and retrieves one BM25 chunk with required terms e and y. A chunk is eligible only if its timestamp is no later than the cutoff. The state returned to the reader is

$$E_t = (O(q),\ C_t,\ n_t,\ closed_t,\ F_t),$$

where C_t is the covered document set, n_t is query count, $closed_t$ is the closure flag, and F_t records future ids observed by diagnostics. The reader is allowed to answer only when $closed_t = 1$; otherwise it can abstain or request more retrieval.

4.3 Complexity and non-oracle contract

For N chunks, BM25 indexing is $O(N)$ in the stored term statistics and each query scans the in-memory chunks. Planning is $O(|O(q)|)$ retrieval steps up to the budget. The planner receives no hidden document ids, answer labels or future gold state. This contract is tested by source inspection and deterministic unit tests.

Table 1: Controlled mechanism benchmark. Higher is better except overclaim, future leakage and queries. Values are means over 5,600 cases.

Method	Acc.	Closure	Overclaim	Counter recall	Queries	Utility	Future leak
Single-shot	.066	.101	.004	.651	10.00	-.407	.000
Fixed decomposition	.175	.293	.014	.658	10.00	-.363	.000
Generic adaptive	.193	.343	.015	.632	10.00	-.371	.000
FinPlan state	.191	.366	.013	.605	8.87	-.358	.000

Table 2: Component ablations under the same 5,600-case contract.

Variant	Closure	Overclaim	Counter recall	Queries	Utility	Future leak
FinPlan state	.366	.013	.605	8.87	-.358	.000
No counterevidence	.511	.035	.315	7.92	-.284	.000
No adaptive stop	.337	.012	.606	10.00	-.370	.000
No point-in-time	.433	.021	.578	8.57	-.438	.502

5 Experiments

5.1 Protocol

We ran the repository benchmark with 20 seeds, 40 cases per scenario, seven scenarios (balanced, multi-path, counterevidence-dense, sparse, source-conflict, future-revision and random-world), and a shared budget of 10. Every method saw the same paths, documents, cutoff and resolver. The utility was fixed before evaluation: correct answer = 1, abstention = -0.15, other error = -1, overclaim penalty = -1, and query cost = -0.03. The full run is identified by SHA-256 `E5D207A2...42CAA28`; its compact summary is committed in `paper/results/controlled_summary.json`.

5.2 Main result and baselines

Table 1 reports means across all case-level records. `single_shot` uses the highest-relevance document, `fixed_decomposition` cycles through a static slot list, and `generic_adaptive` selects the least-confident slot without risk weights. `finplan_state` is the proposed policy.

Relative to single-shot, FinPlan increases closure by 0.2645 and reduces query count by 1.1279. Accuracy is similar to generic adaptive (difference -0.0020), which is an important negative result: obligation state is primarily a process and stopping improvement in this benchmark, not a universal answer-accuracy advantage.

5.3 Ablations and stress tests

Removing counterevidence raises closure (.511) and lowers queries (7.92), but counterevidence recall falls from .605 to .315 and overclaim rises from .013 to .035. Removing adaptive stopping uses all 10 queries and gives utility -.370. Removing temporal filtering creates future leakage in .502 of cases and gives utility -.438. These paired changes isolate the roles of counterevidence, stopping and point-in-time filtering without changing the candidate stream.

5.4 Budget sensitivity

The quality–cost frontier was checked at budgets $1, 2, 4, 6, 10, 16$ with the same 20 seeds and seven scenarios. FinPlan closure increased monotonically from 0.000, 0.000, 0.029, 0.182, 0.366 to 0.455, while counterevidence recall rose from 0.315 to 0.633. Declared utility decreased from -0.180 to -0.451 because the fixed query charge eventually dominated the gain from additional closure. Thus budget 10 is a reproducible operating point, not a universal optimum; the full sensitivity artifact and per-run hashes are in `paper/results/budget_sensitivity.json`.

6 Discussion

The results support H1 only in a bounded sense. Explicit state makes unresolved obligations visible and permits early stopping, while counterevidence and time filters impose measurable costs. The no-counterevidence ablation is a useful challenge to a simplistic "more closure is better" interpretation. The controlled benchmark does not estimate financial answer quality on SEC filings, does not include a learned reader, and does not establish superiority over stronger adaptive or dense retrievers. Closure should therefore be reported alongside correctness, leakage and cost in any future real-data study.

7 Reproducibility and agent interfaces

The package requires Python 3.10 and NumPy; 'python -m pytest -q' runs the contract tests. 'finplan-rag-agent –schema' emits an OpenAI-compatible tool schema. The default command reads one JSON request per line and emits one JSON response per line, so a Codex or Claude Code tool wrapper can call it without vendor-specific state. The response includes obligations, covered document ids, selected evidence chunks, closure, future ids, query count and a conservative recommendation to answer or abstain. It never fabricates a reader answer.

8 Conclusion

FinPlan-RAG treats financial retrieval as obligation completion under a cutoff, not only ranking. The theory identifies when a score-only stopping rule is insufficient; the controlled evidence shows the measurable trade-off introduced by explicit counterevidence and temporal filtering. The next decisive step is a frozen, independently scored financial filing benchmark with a shared reader, strong retrieval baselines and held-out point-in-time splits.

A Implementation contract

The public data structures are `DocumentObligation`, `Chunk`, `RetrievalResult` and `EvidenceState`. `EvidenceState.closed` is true only when every obligation with a known document id is covered. `future_document_ids` is diagnostic and is kept separate from the consumed point-in-time context. Reader output is checked by deterministic numeric support and refusal parsing.

B Evidence boundary

The summary values in Table 1 trace to the local JSON source hash listed above. They are not rederived from prose, and no external SQCAD result is presented as a FinPlan-RAG result. Real-data results, model weights and large corpora remain external under the repository storage contract.

References

Lewis, P. et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS* (2020).

Maharana, A. et al. LoCoMo: Long-context conversation memory benchmark. *ACL* (2024).

Wu, L. et al. LongMemEval: Benchmarking long-term memory of large language models. arXiv (2024).